

Docker Persistent Storage (Volumes) - Complete Demonstration

1. Concept

Containers are temporary. If you delete a container, all data inside it is lost. Docker volumes help store data permanently outside containers.

2. Without Volume (Data Loss)

```
docker run -it --name c1 ubuntu
cd /tmp
echo "Hello Students" > file.txt
cat file.txt
exit
docker rm c1
docker run -it --name c2 ubuntu
cd /tmp
ls
```

Result: file.txt is lost.

3. With Volume (Persistent Data)

```
docker volume create mydata
docker run -it --name c3 -v mydata:/app ubuntu
cd /app
echo "Persistent Data" > data.txt
cat data.txt
exit
docker rm c3
docker run -it --name c4 -v mydata:/app ubuntu
cd /app
ls
cat data.txt
```

Result: Data is preserved even after container deletion.

4. Key Difference

Without volume: Data lost. With volume: Data persists.

5. Common Errors

- Using different volume names
- Wrong mount path
- Volume still attached to container

6. Scenario Question

A company loses uploaded files after restarting containers. Solution: Use Docker volumes to persist data.

